Data Exploration

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Load in Packages

```r
#load in required packages
library(tidyr)
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
library(ggplot2)
library(patchwork)
library(brms)
```

```
Loading required package: Rcpp

Loading 'brms' package (version 2.22.0). Useful instructions
can be found by typing help('brms'). A more detailed introduction
to the package is available through vignette('brms_overview').
```

```
Attaching package: 'brms'

The following object is masked from 'package:stats':

    ar
```

```r
library(ggeffects)
```

Unzip and Read in Files

Unzip

```r
#unzip rodent activity file
# unzip("edi.2200.1.zip")
```

Read in files

```r
#read in rodent activity file
rodent_data <- read.csv("data/Rodent_activity.csv")
```

```r
#read in trapping data file
trapping_data <- read.csv("data/Trapping.csv")
```

```r
#read in vegetation cover data file
vegetation_data <- read.csv("data/Vegetation_cover.csv")
```

```r
#read in seed predation data
seed_predation_data <- read.csv("data/Seed_predation.csv")
```

Data Wrangling

Separate

```r
#separate data in trapping_data
colnames(trapping_data) #obtain column names to know what to call new columns
```

```
[1] "Trap_day.Lynx.Habitat.Plot.Trap.Species.Working"
```

```r
#use separate to separate on '.'
trapping_separated <- trapping_data %>%
  separate(col = Trap_day.Lynx.Habitat.Plot.Trap.Species.Working,
           into = c("Trap_day", "Lynx", "Habitat", "Plot", "Trap", "Species", "Working"),
           sep = ";")
```

```r
#separate data in vegetation_data
colnames(vegetation_data)
```

```
[1] "Lynx.Plot.Habitat.Tree.Shrub.Rock"
```

```r
vegetation_separated <- vegetation_data %>%
  separate(col = Lynx.Plot.Habitat.Tree.Shrub.Rock,
           into = c("Lynx", "Plot", "Habitat", "Tree", "Shrub", "Rock"),
           sep = ";")
```

```r
#separate seed predation data
colnames(seed_predation_data)
```

```
[1] "Lynx.Habitat.Plot.Microhabitat.Code.CAM.Seeds_left0.Seeds_left1.Seeds_predated1.Seeds_le
```

```r
seed_predation_separated <- seed_predation_data %>%
  separate(col = Lynx.Habitat.Plot.Microhabitat.Code.CAM.Seeds_left0.Seeds_left1.Seeds_predat
           into = c("Lynx", "Habitat", "Plot", "Microhabitat", "Code", "CAM", "Seeds_left0",
           sep = ";")
```

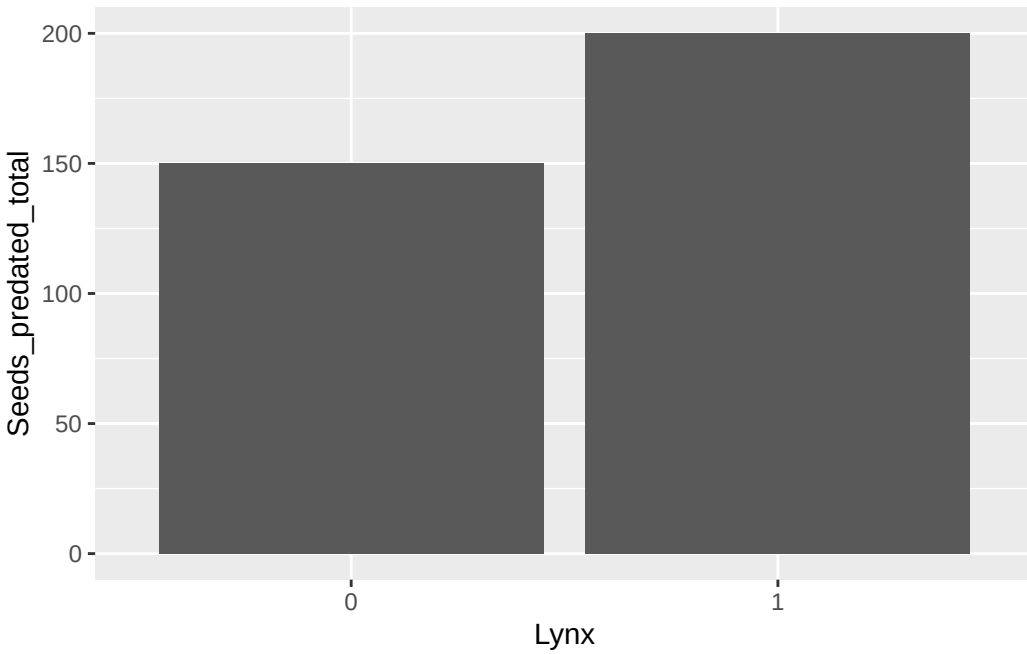
As.Numeric

```r
#make total seeds predated column and seeds left 0 column numeric
#Seeds_left0 is used as total seeds left out at the beginning (8 seeds)

seed_predation_separated <- seed_predation_separated %>%
  mutate(Seeds_predated_total = as.numeric(Seeds_predated_total),
         Seeds_left0 = as.numeric(Seeds_left0))
```

Graphs!

3 Simple Bar Graphs

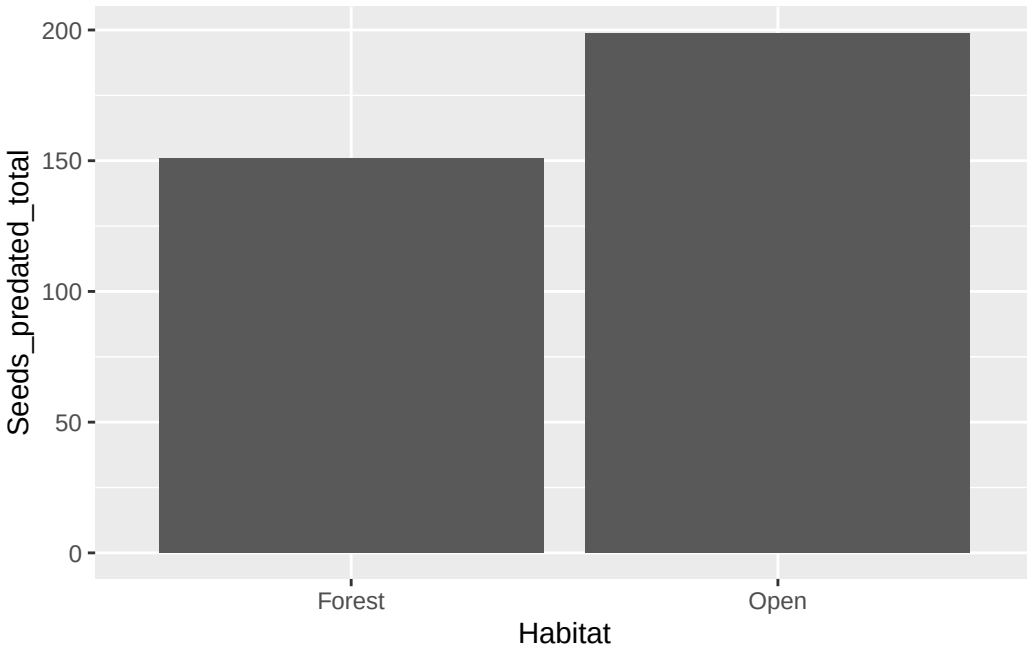
```
#plotting seed predation total as a function of lynx presence
ggplot(data = seed_predation_separated, aes(x = Lynx, y = Seeds_predated_total)) +
  geom_col()
```




```
ggsave("output/SeedPredationLynxPresence.png") #save to output folder
```

```
Saving 5.5 x 3.5 in image
```

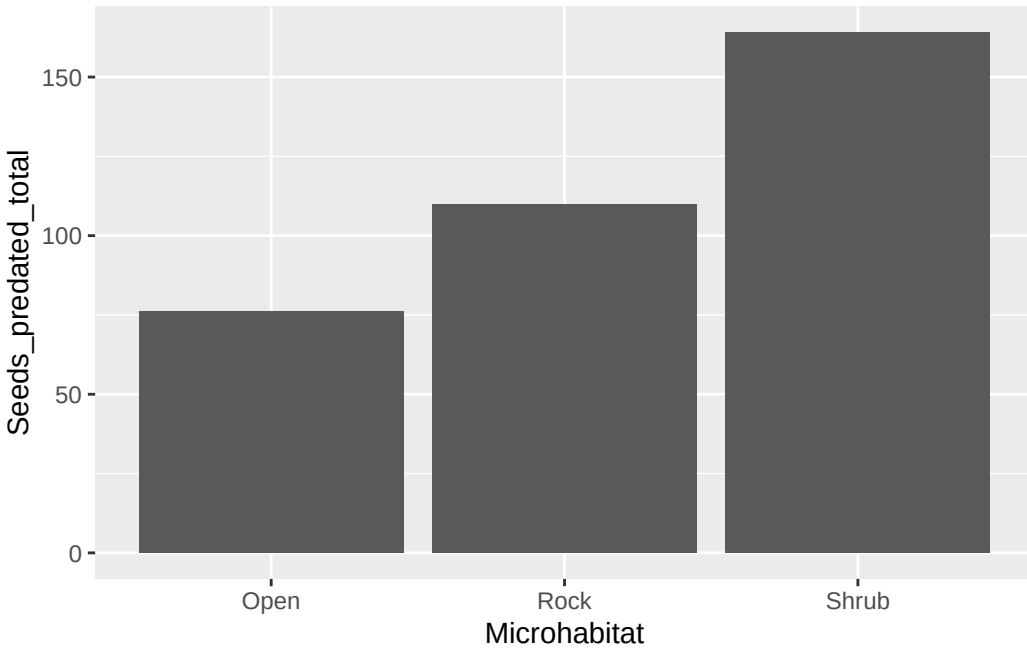
```
#plotting seed predation total as a function of macrohabitat type
ggplot(data = seed_predation_separated, aes(x = Habitat, y = Seeds_predated_total)) +
  geom_col()
```




```
ggsave("output/SeedPredationMacrohabitat.png")
```

```
Saving 5.5 x 3.5 in image
```

```
#plotting seed predation total as a function of microhabitat type
ggplot(data = seed_predation_separated, aes(x = Microhabitat, y = Seeds_predated_total)) +
  geom_col()
```




```
ggsave("output/SeedPredationMicrohabitat.png")
```

```
Saving 5.5 x 3.5 in image
```

6 Bar Graphs with All Variables

```
#graph 1: lynx = 1, microhabitat = open

#make new data frame with just rows where lynx is present AND microhabitat is open
lynx_open <- filter(seed_predation_separated, Lynx == "1", Microhabitat == "Open")

g1 <- ggplot(data = lynx_open, aes(x = Habitat, y = Seeds_predated_total, fill = Habitat)) +
  geom_col() +
  labs(x = "Habitat",
       y = "Total Number of Seeds Predated",
       title = "Lynx Present") +
  ylim(0, 50)
```

```
#graph 2: lynx = 1, microhabitat = rock
```

```r
#make new data frame with just rows where lynx is present AND microhabitat is rock
lynx_rock <- filter(seed_predation_separated, Lynx == "1", Microhabitat == "Rock")

g2 <- ggplot(data = lynx_rock, aes(x = Habitat, y = Seeds_predated_total, fill = Habitat)) +
  geom_col() +
  labs(x = "Habitat",
       y = "Total Number of Seeds Predated",
       title = "Lynx Present") +
  ylim(0, 50)
```

```r
#graph 3: lynx = 1, microhabitat = shrub

#make new data frame with just rows where lynx is present AND microhabitat is shrub
lynx_shrub <- filter(seed_predation_separated, Lynx == "1", Microhabitat == "Shrub")

g3 <- ggplot(data = lynx_shrub, aes(x = Habitat, y = Seeds_predated_total, fill = Habitat)) +
  geom_col() +
  labs(x = "Habitat",
       y = "Total Number of Seeds Predated",
       title = "Lynx Present") +
  ylim(0, 50)
```

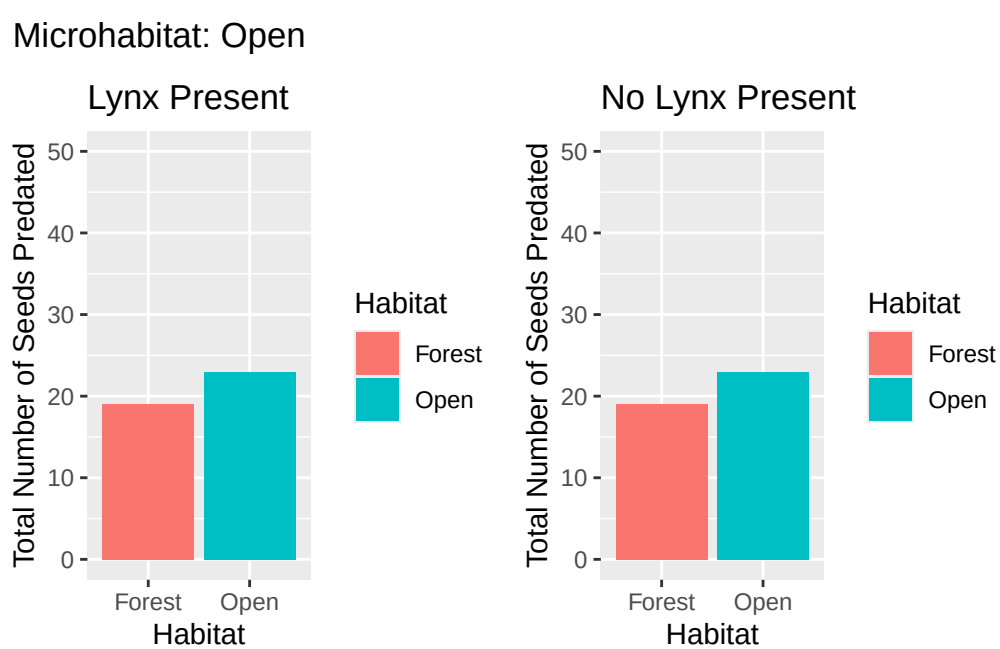
```r
#graph 4: lynx = 0, microhabitat = open

#make new data frame with just rows where lynx is not present AND microhabitat is open
noLynx_open <- filter(seed_predation_separated, Lynx == "0", Microhabitat == "Open")

g4 <- ggplot(data = lynx_open, aes(x = Habitat, y = Seeds_predated_total, fill = Habitat)) +
  geom_col() +
  labs(x = "Habitat",
       y = "Total Number of Seeds Predated",
       title = "No Lynx Present") +
  ylim(0, 50)
```

```r
#graph 5: lynx = 0, microhabitat = rock

#make new data frame with just rows where lynx is not present AND microhabitat is rock
noLynx_rock <- filter(seed_predation_separated, Lynx == "0", Microhabitat == "Rock")

g5 <- ggplot(data = noLynx_rock, aes(x = Habitat, y = Seeds_predated_total, fill = Habitat))
  geom_col() +
  labs(x = "Habitat",
```

```
       y = "Total Number of Seeds Predated",
       title = "No Lynx Present") +
  ylim(0, 50)
```

```
#graph 6: lynx = 0, microhabitat = shrub

#make new data frame with just rows where lynx is not present AND microhabitat is shrub
noLynx_shrub <- filter(seed_predation_separated, Lynx == "0", Microhabitat == "Shrub")

g6 <- ggplot(data = noLynx_shrub, aes(x = Habitat, y = Seeds_predated_total, fill = Habitat))
  geom_col() +
  labs(x = "Habitat",
       y = "Total Number of Seeds Predated",
       title = "No Lynx Present") +
  ylim(0, 50)
```

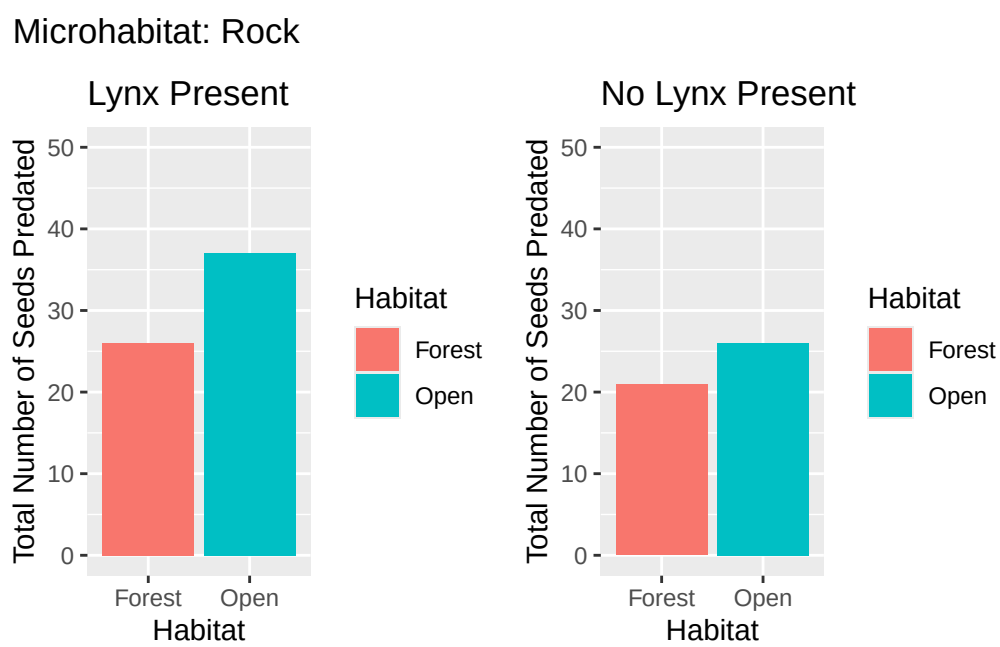
```
#put the two graphs with the open microhabitat type together
(g1 + g4) + plot_annotation(
  title = 'Microhabitat: Open')
```



```r
ggsave("output/SeedPredationLynxPresenceMacrohabitatMicrohabitatOpen.png")
```

```
Saving 5.5 x 3.5 in image
```

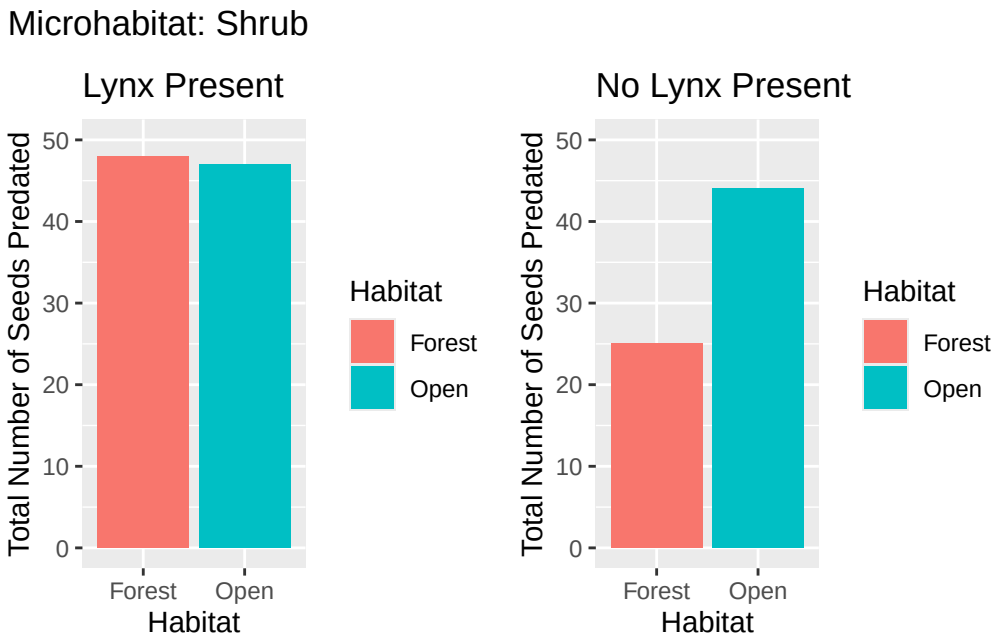
```r
##put the two graphs with the rock microhabitat type together
(g2 + g5) + plot_annotation(
  title = 'Microhabitat: Rock')
```




```r
ggsave("output/SeedPredationLynxPresenceMacrohabitatMicrohabitatRock.png")
```

```
Saving 5.5 x 3.5 in image
```

```r
#put the two graphs with the shrub microhabitat type together
(g3 + g6) + plot_annotation(
  title = 'Microhabitat: Shrub')
```

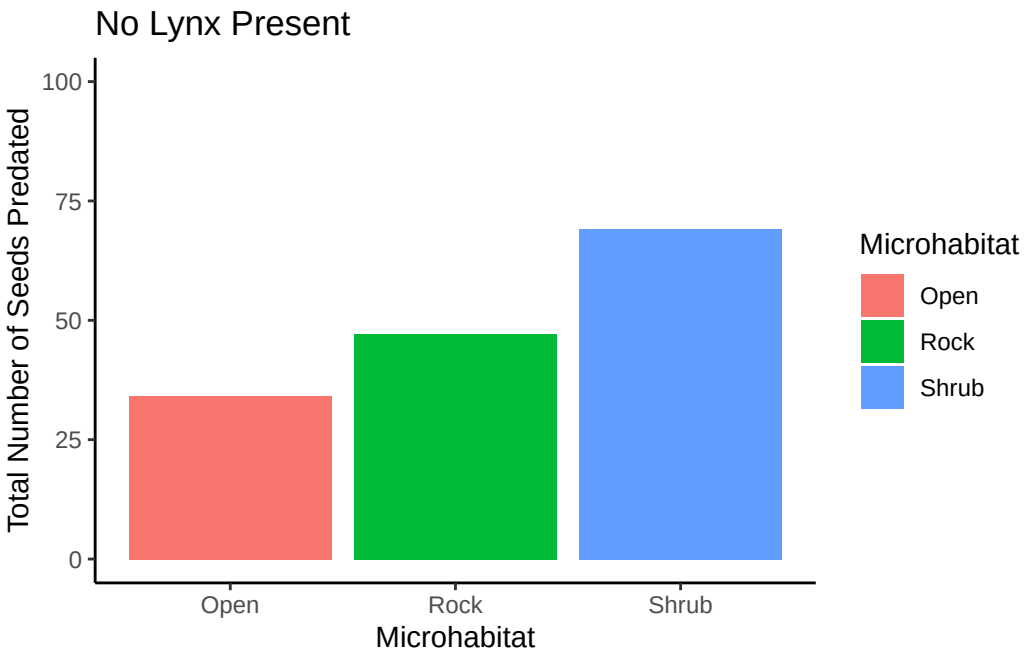



```
ggsave("output/SeedPredationLynxPresenceMacrohabitatMicrohabitatShrub.png")
```

```
Saving 5.5 x 3.5 in image
```

2 Bar Graphs of Seeds Predated Against Microhabitat Type

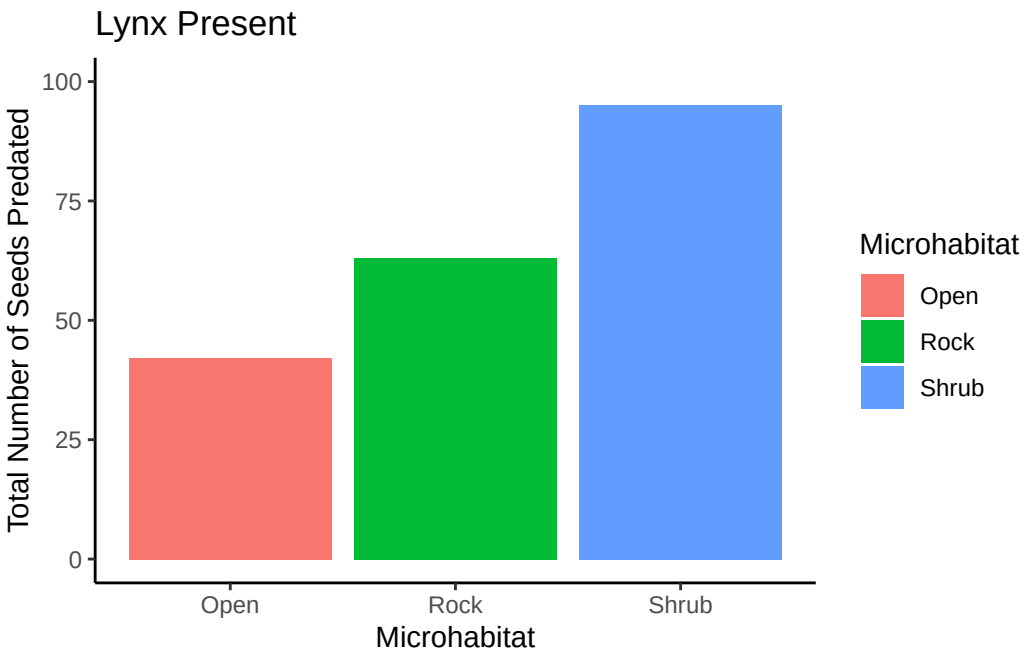
```
#create a new data frame where all rows have no lynx present
lynx_absent <- filter(seed_predation_separated, Lynx == "0")

#graph seeds predated as a function of microhabitat type
ggplot(data = lynx_absent, aes(x = Microhabitat, y = Seeds_predated_total, fill = Microhabita
  geom_col() +
  labs(x = "Microhabitat",
       y = "Total Number of Seeds Predated",
       title = "No Lynx Present") +
  ylim(0, 100) +
  theme_classic()
```




```
ggsave("output/SeedPredationNoLynxMicrohabitat.png")
```

```
Saving 5.5 x 3.5 in image
```

```
#create a new data frame where all rows have lynx present
lynx_present <- filter(seed_predation_separated, Lynx == "1")

#graph seeds predated as a function of microhabitat type
ggplot(data = lynx_present, aes(x = Microhabitat, y = Seeds_predated_total, fill = Microhabi
  geom_col() +
  labs(x = "Microhabitat",
       y = "Total Number of Seeds Predated",
       title = "Lynx Present") +
  ylim(0, 100) +
  theme_classic()
```

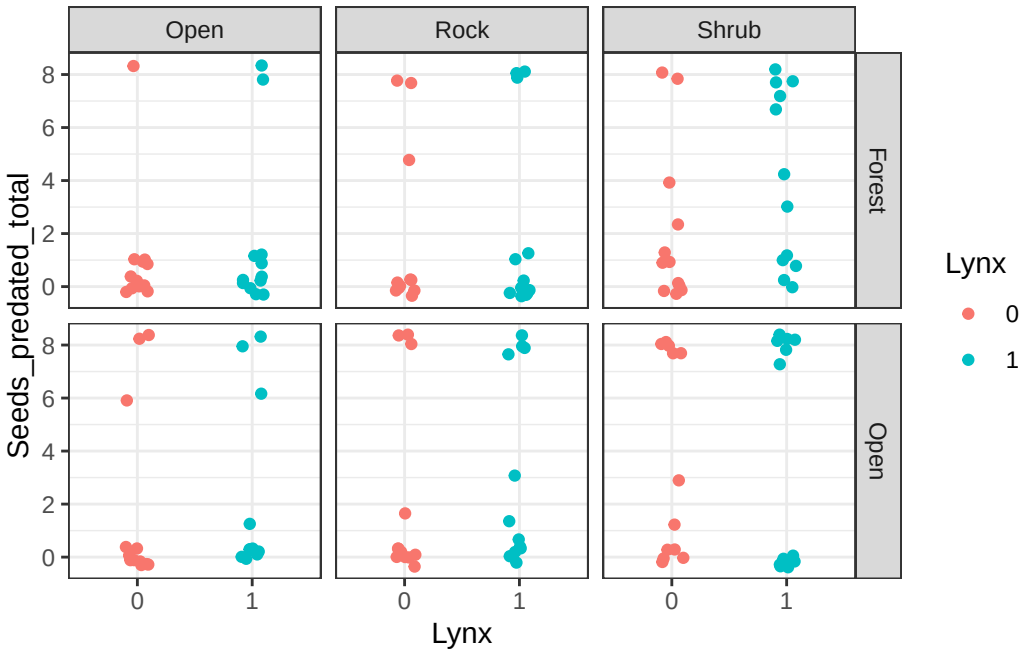



```
ggsave("output/SeedPredationLynxPresentMicrohabitat.png")
```

```
Saving 5.5 x 3.5 in image
```

Jitter Plot with all 4 Variables Represented

```
#Graph seeds predated in 6 panels representing the different combinations of experimental co

seed_predation_separated %>%
ggplot(aes(x = Lynx, y = Seeds_predated_total, color = Lynx)) +
  geom_jitter(width = 0.1) +
  # labs(x = "Microhabitat",
  #      y = "Total Number of Seeds Predated",
  #      title = "Lynx Present") +
  facet_grid(Habitat ~ Microhabitat) +
  theme_bw()
```




```
ggsave("output/SeedPredationLynxMacrohabitatMicrohabitat.png")
```

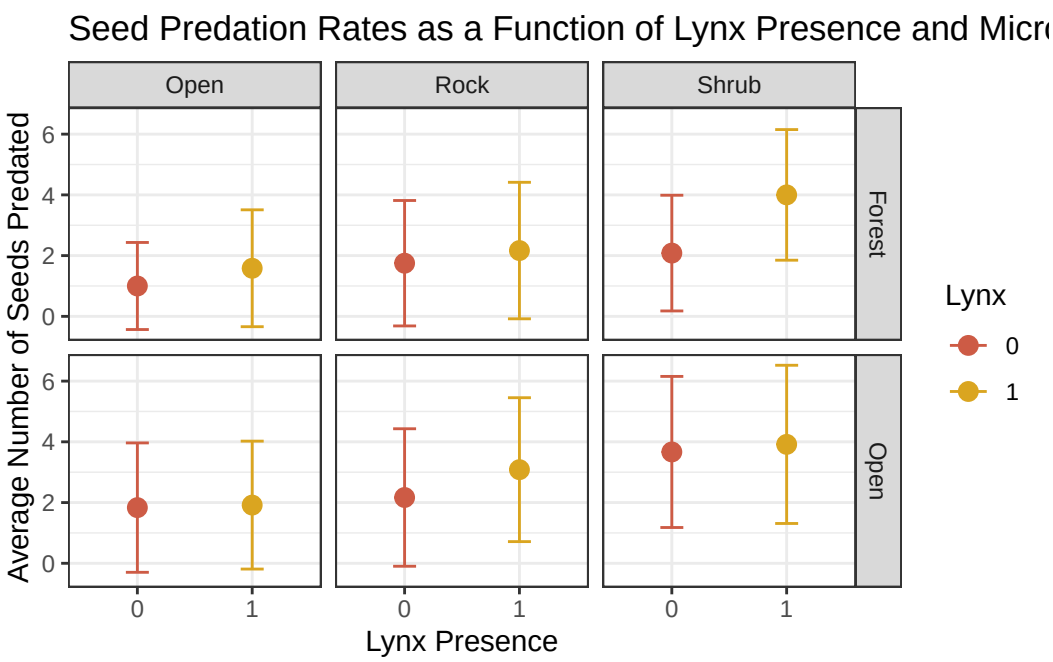
```
Saving 5.5 x 3.5 in image
```

Final Visualization

```
# Use these commands to find average number of seeds predated per experimental condition comb
summary_stats<- seed_predation_separated %>% group_by(Lynx, Habitat, Microhabitat) %>%summari
   mean_seeds = mean(Seeds_predated_total, na.rm = TRUE),
    n = sum(!is.na(Seeds_predated_total)),
    sd_seeds = sd(Seeds_predated_total, na.rm = TRUE),
    se_seeds = sd_seeds / sqrt(n),
    ci_lower = mean_seeds - qt(0.975, df = n - 1) * se_seeds,
    ci_upper = mean_seeds + qt(0.975, df = n - 1) * se_seeds
   )%>%
  ungroup()
```

```
`summarise()` has grouped output by 'Lynx', 'Habitat'. You can override using
the `.groups` argument.
```

```
#Graph the average number of seeds predated per each experimental condition combiantion
ggplot(summary_stats, aes(x = Lynx, y = mean_seeds, color = Lynx)) +
  geom_point(size = 3) +  # Average points
  geom_errorbar(aes(ymin = ci_lower, ymax = ci_upper), width = 0.2) +  # 95% CI error bars
  facet_grid(Habitat ~ Microhabitat) +
  labs(
    x = "Lynx Presence",
    y = "Average Number of Seeds Predated",
    title = "Seed Predation Rates as a Function of Lynx Presence and Micro/Macro habitat") +
  scale_color_manual(values = c("1" = "goldenrod", "0" = "coral3")) +
  theme_bw()
```



Models

Additive Model

```
#Count the number of trials for each combination of experimental variables (just to see how m
seed_predation_separated %>%
  count(Lynx, Habitat, Microhabitat, Plot)
```

	Lynx	Habitat	Microhabitat	Plot	n
1	0	Forest	Open	1CSL	2
2	0	Forest	Open	2CSL	2
3	0	Forest	Open	3CSL	2
4	0	Forest	Open	4CSL	2
5	0	Forest	Open	5CSL	2
6	0	Forest	Open	6CSL	2
7	0	Forest	Rock	1CSL	2
8	0	Forest	Rock	2CSL	2
9	0	Forest	Rock	3CSL	2
10	0	Forest	Rock	4CSL	2
11	0	Forest	Rock	5CSL	2
12	0	Forest	Rock	6CSL	2
13	0	Forest	Shrub	1CSL	2
14	0	Forest	Shrub	2CSL	2
15	0	Forest	Shrub	3CSL	2
16	0	Forest	Shrub	4CSL	2
17	0	Forest	Shrub	5CSL	2
18	0	Forest	Shrub	6CSL	2
19	0	Open	Open	1OSL	2
20	0	Open	Open	2OSL	2
21	0	Open	Open	3OSL	2
22	0	Open	Open	4OSL	2
23	0	Open	Open	5OSL	2
24	0	Open	Open	6OSL	2
25	0	Open	Rock	1OSL	2
26	0	Open	Rock	2OSL	2
27	0	Open	Rock	3OSL	2
28	0	Open	Rock	4OSL	2
29	0	Open	Rock	5OSL	2
30	0	Open	Rock	6OSL	2
31	0	Open	Shrub	1OSL	2
32	0	Open	Shrub	2OSL	2
33	0	Open	Shrub	3OSL	2
34	0	Open	Shrub	4OSL	2
35	0	Open	Shrub	5OSL	2
36	0	Open	Shrub	6OSL	2
37	1	Forest	Open	1CL	2
38	1	Forest	Open	2CL	2
39	1	Forest	Open	3CL	2
40	1	Forest	Open	4CL	2
41	1	Forest	Open	5CL	2
42	1	Forest	Open	6CL	2

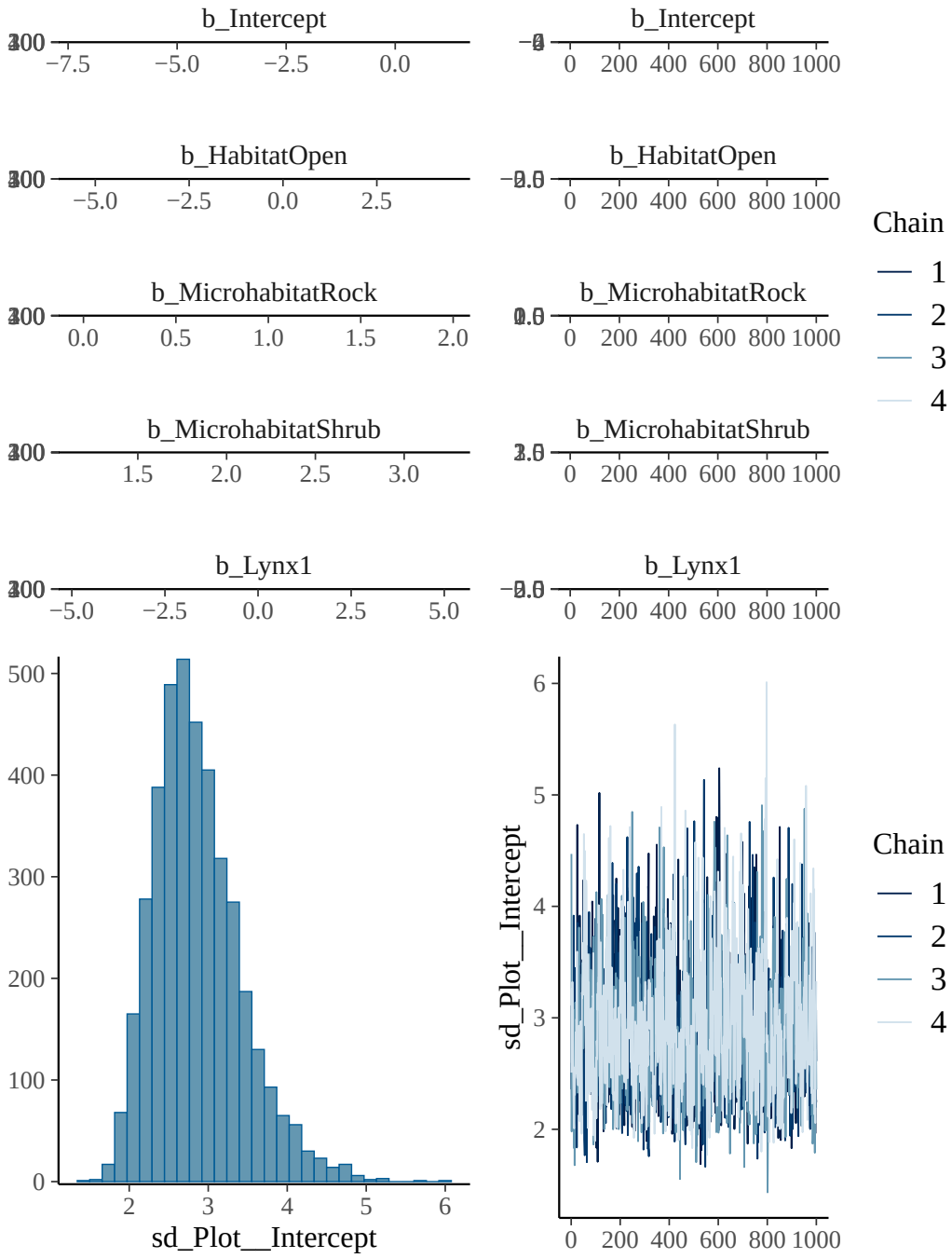
```
43    1  Forest         Rock  1CL 2
44    1  Forest         Rock  2CL 2
45    1  Forest         Rock  3CL 2
46    1  Forest         Rock  4CL 2
47    1  Forest         Rock  5CL 2
48    1  Forest         Rock  6CL 2
49    1  Forest        Shrub  1CL 2
50    1  Forest        Shrub  2CL 2
51    1  Forest        Shrub  3CL 2
52    1  Forest        Shrub  4CL 2
53    1  Forest        Shrub  5CL 2
54    1  Forest        Shrub  6CL 2
55    1    Open         Open  1OL 2
56    1    Open         Open  2OL 2
57    1    Open         Open  3OL 2
58    1    Open         Open  4OL 2
59    1    Open         Open  5OL 2
60    1    Open         Open  6OL 2
61    1    Open         Rock  1OL 2
62    1    Open         Rock  2OL 2
63    1    Open         Rock  3OL 2
64    1    Open         Rock  4OL 2
65    1    Open         Rock  5OL 2
66    1    Open         Rock  6OL 2
67    1    Open        Shrub  1OL 2
68    1    Open        Shrub  2OL 2
69    1    Open        Shrub  3OL 2
70    1    Open        Shrub  4OL 2
71    1    Open        Shrub  5OL 2
72    1    Open        Shrub  6OL 2
```

```r
#Run an addtive model of number of seeds predated ~ Microhabitat + Lynx Presence
m.seeds.habitat.microhabitat.lynx <-
brm(data = seed_predation_separated, # Give the model the data
# Choose a binomial distribution --> There is a set number (8 seeds) and it is success vs. fa
family = binomial(link = "logit"),
# Specify the model here.
Seeds_predated_total | trials(Seeds_left0) ~ 1 + Habitat + Microhabitat + Lynx + (1|Plot), #t
# Specify parameters for executing the Markov chains
iter = 2000, warmup = 1000, chains = 4, cores = 4,
# Save the model in output folder
file = "output/m.seeds.habitat.microhabitat.lynx")
```

```r
#get summary of the additive model
summary(m.seeds.habitat.microhabitat.lynx)
```

```
 Family: binomial
  Links: mu = logit
Formula: Seeds_predated_total | trials(Seeds_left0) ~ 1 + Habitat + Microhabitat + Lynx + (1
   Data: seed_predation_separated (Number of observations: 144)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~Plot (Number of levels: 24)
              Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)     2.87      0.56     1.99     4.18 1.00      976     1899

Regression Coefficients:
                  Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
Intercept            -3.06      1.06    -5.18    -1.06 1.00     1503     2034
HabitatOpen           0.16      1.18    -2.16     2.48 1.00     1517     1936
MicrohabitatRock      0.94      0.25     0.46     1.44 1.00     3030     2391
MicrohabitatShrub     2.19      0.26     1.71     2.72 1.00     2937     2756
Lynx1                 0.92      1.22    -1.42     3.47 1.00     1429     1493

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
#check that model ran correctly
plot(m.seeds.habitat.microhabitat.lynx)
```


b_Intercept
b_HabitatOpen
b_MicrohabitatRock
b_MicrohabitatShrub
b_Lynx1
sd_Plot__Intercept
Chain
1
2
3
4

Interactive Model

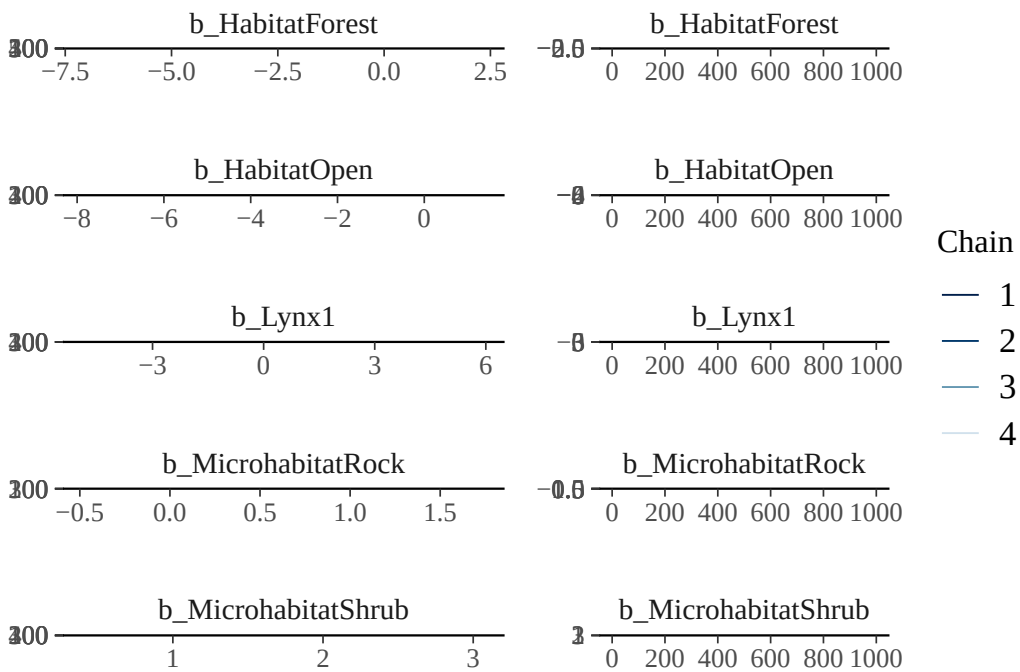
```r
# Run an interactive model of number of seeds predated ~ Microhabitat + Lynx Presence
m.seeds.habitat.microhabitat.lynx.int <-
brm(data = seed_predation_separated, # Give the model the data
family = binomial(link = "logit"),
# Specify the model here.
Seeds_predated_total | trials(Seeds_left0) ~ 0 + Habitat + Lynx*Microhabitat + (1|Plot),
# Specify parameters for executing the Markov chains
iter = 2000, warmup = 1000, chains = 4, cores = 4,
file = "output/m.seeds.habitat.microhabitat.lynx.int")
```

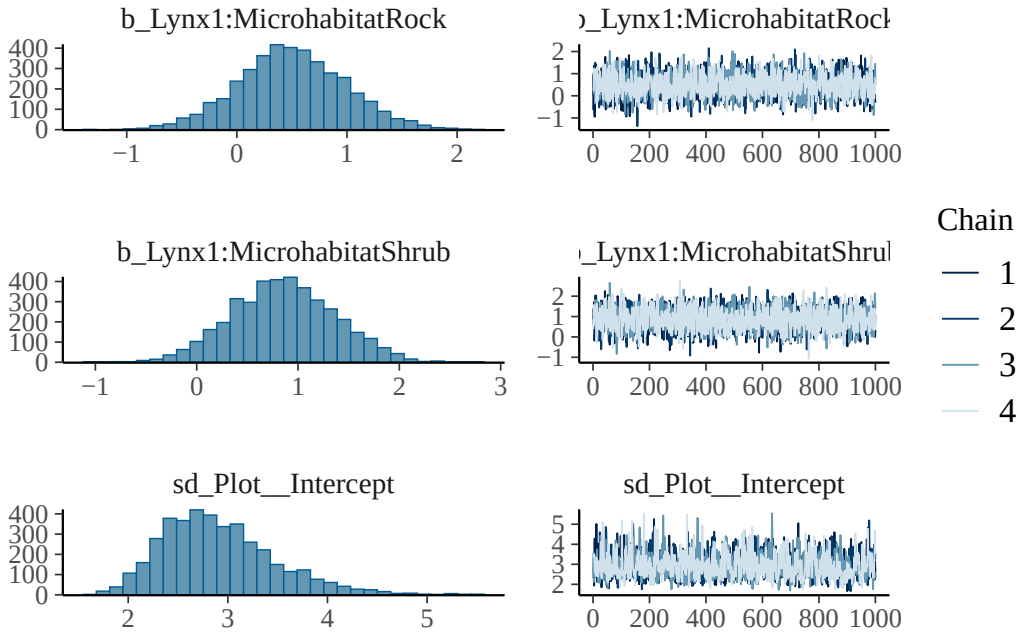
```r
#get summary of the interactive model
summary(m.seeds.habitat.microhabitat.lynx.int)
```

```
 Family: binomial
  Links: mu = logit
Formula: Seeds_predated_total | trials(Seeds_left0) ~ 0 + Habitat + Lynx * Microhabitat + (1
   Data: seed_predation_separated (Number of observations: 144)
  Draws: 4 chains, each with iter = 2000; warmup = 1000; thin = 1;
         total post-warmup draws = 4000

Multilevel Hyperparameters:
~Plot (Number of levels: 24)
              Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
sd(Intercept)     2.91      0.57     2.01     4.23 1.00     1100     1886

Regression Coefficients:
                        Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS
HabitatForest              -2.91      1.13    -5.24    -0.69 1.00     1308
HabitatOpen                -2.77      1.14    -5.05    -0.65 1.00     1504
Lynx1                       0.39      1.32    -2.24     3.02 1.00     1331
MicrohabitatRock            0.72      0.34     0.04     1.39 1.00     2741
MicrohabitatShrub           1.79      0.35     1.11     2.49 1.00     2773
Lynx1:MicrohabitatRock      0.50      0.49    -0.45     1.49 1.00     2917
Lynx1:MicrohabitatShrub     0.86      0.51    -0.12     1.86 1.00     3133
                        Tail_ESS
HabitatForest               1380
HabitatOpen                 1939
Lynx1                       1636
MicrohabitatRock            2959
```

```
MicrohabitatShrub           2566
Lynx1:MicrohabitatRock      2635
Lynx1:MicrohabitatShrub     3064

Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
and Tail_ESS are effective sample size measures, and Rhat is the potential
scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
#check that model ran correctly
plot(m.seeds.habitat.microhabitat.lynx.int)
```





Model Summary

Additive Model:

Intercept = Forest Habitat, Open Microhabitat, No Lynx

Open habitat: 0.16 increase in seeds predated compared to forest (not reasonably different from 0)

Rock Microhabitat: 0.94 increase in seeds predated compared to open (reasonably different from 0)

Shrub Microhabitat: 2.19 increase in seeds predated compared to open (reasonably different from 0)

Lynx Present: 0.92 increase in seeds predated compared to lynx absent (not reasonably different from 0)

```
#check PSIS value
loo(m.seeds.habitat.microhabitat.lynx)
```

```
Warning: Found 17 observations with a pareto_k > 0.7 in model
'm.seeds.habitat.microhabitat.lynx'. We recommend to set 'moment_match = TRUE'
```

```
in order to perform moment matching for problematic observations.


Computed from 4000 by 144 log-likelihood matrix.

         Estimate   SE
elpd_loo   -365.8 44.2
p_loo       107.6 16.0
looic       731.7 88.3
------
MCSE of elpd_loo is NA.
MCSE and ESS estimates assume MCMC draws (r_eff in [0.5, 1.4]).

Pareto k diagnostic values:
                         Count Pct.    Min. ESS
(-Inf, 0.7]   (good)     127   88.2%   110
   (0.7, 1]   (bad)       13    9.0%   <NA>
   (1, Inf)   (very bad)   4    2.8%   <NA>
See help('pareto-k-diagnostic') for details.
```

```r
#check WAIC value
waic(m.seeds.habitat.microhabitat.lynx)
```

```
Warning:
49 (34.0%) p_waic estimates greater than 0.4. We recommend trying loo instead.


Computed from 4000 by 144 log-likelihood matrix.

          Estimate   SE
elpd_waic   -359.9 43.8
p_waic       101.7 15.5
waic         719.9 87.5

49 (34.0%) p_waic estimates greater than 0.4. We recommend trying loo instead.
```

Interactive Model:

Difference from 0

Forest (-2.91) and open (-2.77) habitats are reasonably different from 0.

In rock microhabitat, presence of lynx is not reasonably different from 0. This means that presence or absence of lynx does not significantly alter seed predation success in rock microhabitat.

In shrub microhabitat, presence of lynx is not reasonably different from 0. This means that presence or absence of lynx does not significantly alter seed predation success in shrub microhabitat. Approaching on 0, so there is more of a difference between predator presence or absence than in rock microhabitat.

```
#check PSIS value
loo(m.seeds.habitat.microhabitat.lynx.int)
```

```
Warning: Found 22 observations with a pareto_k > 0.7 in model
'm.seeds.habitat.microhabitat.lynx.int'. We recommend to set 'moment_match =
TRUE' in order to perform moment matching for problematic observations.



Computed from 4000 by 144 log-likelihood matrix.

         Estimate   SE
elpd_loo   -375.8 46.6
p_loo       121.9 19.0
looic       751.6 93.1
------
MCSE of elpd_loo is NA.
MCSE and ESS estimates assume MCMC draws (r_eff in [0.4, 1.3]).

Pareto k diagnostic values:
                         Count Pct.    Min. ESS
(-Inf, 0.7]   (good)     122   84.7%   159
   (0.7, 1]   (bad)       16   11.1%   <NA>
   (1, Inf)   (very bad)   6    4.2%   <NA>
See help('pareto-k-diagnostic') for details.
```

```
#check WAIC value
waic(m.seeds.habitat.microhabitat.lynx.int)
```

```
Warning:
48 (33.3%) p_waic estimates greater than 0.4. We recommend trying loo instead.
```

```
Computed from 4000 by 144 log-likelihood matrix.

          Estimate   SE
elpd_waic   -369.6 46.2
p_waic       115.7 18.5
waic         739.3 92.5

48 (33.3%) p_waic estimates greater than 0.4. We recommend trying loo instead.
```

Predictive Plots

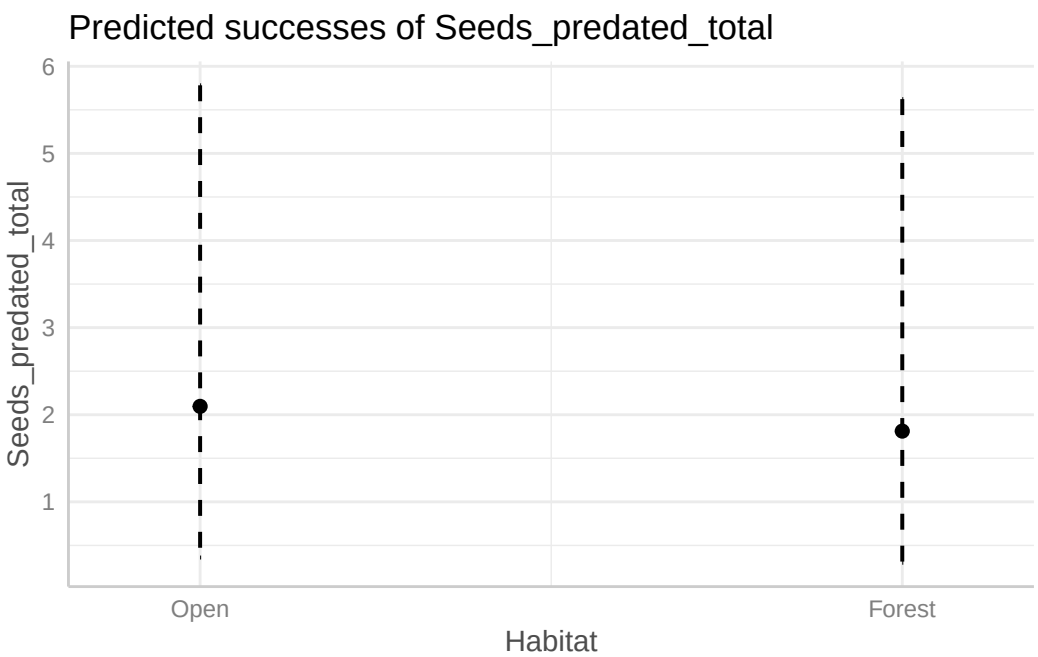
```r
# run a predictive plot to see help evaluate model correctness
pred.add <- predict_response(m.seeds.habitat.microhabitat.lynx, condition = c(Seeds_left0 = 8
```

```
You are calculating adjusted predictions on the population-level (i.e.
  `type = "fixed"`) for a *generalized* linear mixed model.
  This may produce biased estimates due to Jensen's inequality. Consider
  setting `bias_correction = TRUE` to correct for this bias.
  See also the documentation of the `bias_correction` argument.

Some of the focal terms are of type `character`. This may lead to
  unexpected results. It is recommended to convert these variables to
  factors before fitting the model.
  The following variables are of type character: `Habitat`

Note: uncertainty of error terms are not taken into account. Consider
  setting `interval` to "prediction". This will call `posterior_predict()`
  instead of `posterior_epred()`.

Some of the focal terms are of type `character`. This may lead to
  unexpected results. It is recommended to convert these variables to
  factors before fitting the model.
  The following variables are of type character: `Microhabitat`

Note: uncertainty of error terms are not taken into account. Consider
  setting `interval` to "prediction". This will call `posterior_predict()`
  instead of `posterior_epred()`.
```

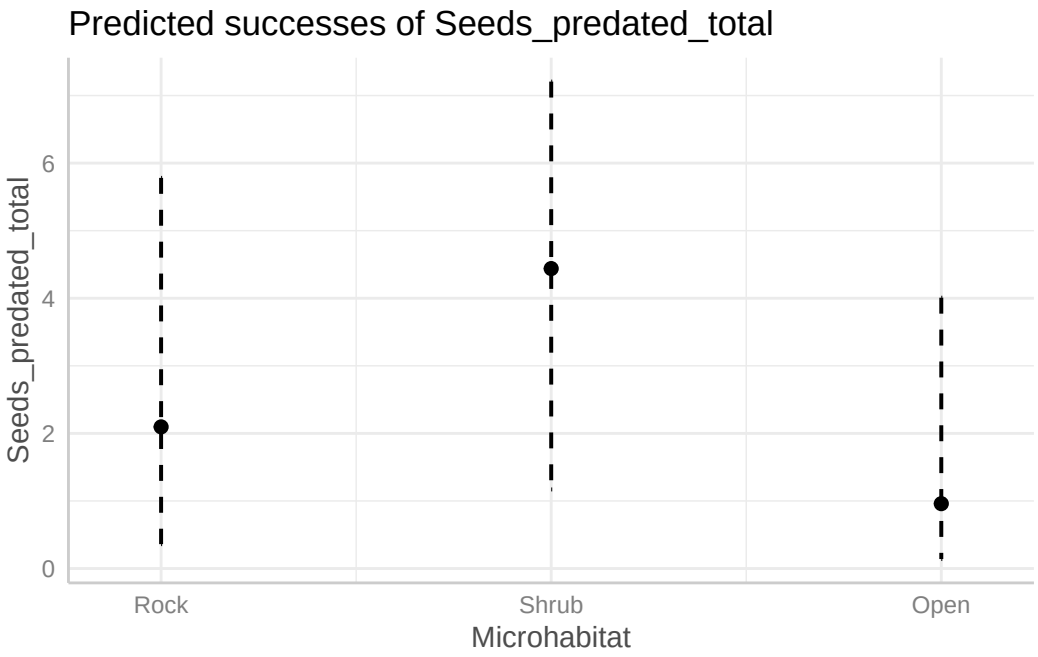
```
Some of the focal terms are of type `character`. This may lead to
  unexpected results. It is recommended to convert these variables to
  factors before fitting the model.
  The following variables are of type character: `Lynx`

Note: uncertainty of error terms are not taken into account. Consider
  setting `interval` to "prediction". This will call `posterior_predict()`
  instead of `posterior_epred()`.
```

```
plot(pred.add)
```

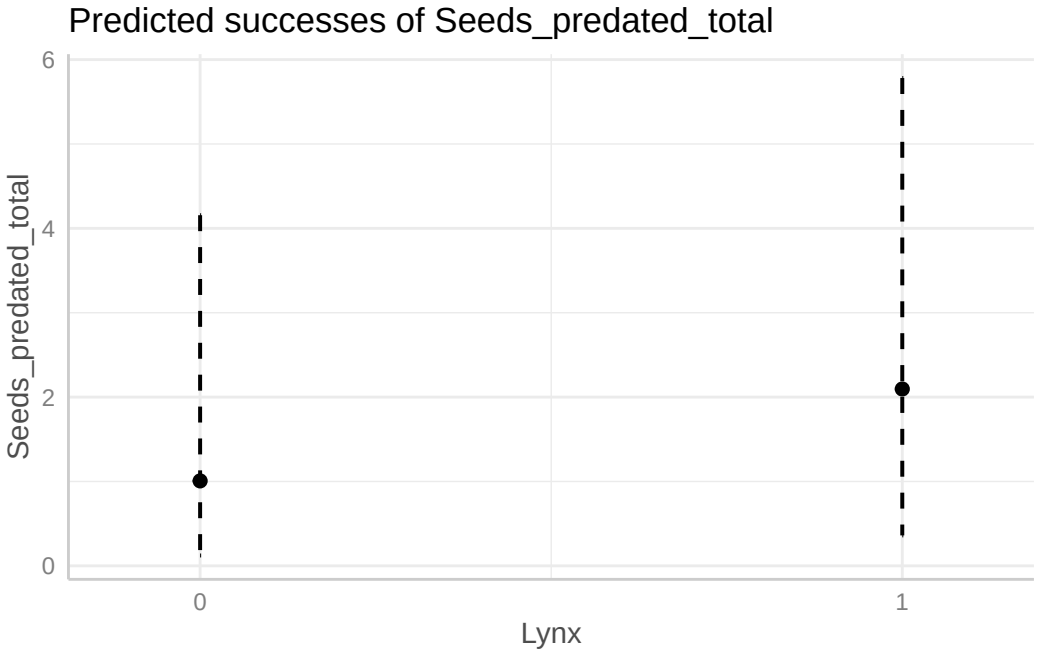
```
$Habitat
```




```
$Microhabitat
```


Predicted successes of Seeds_predated_total
Seeds_predated_total
0
2
4
6
Rock
Shrub
Open
Microhabitat

```
$Lynx
```


Predicted successes of Seeds_predated_total
Seeds_predated_total
0
2
4
6
0
1
Lynx

Compare Models

The additive model is better because it has lower PSIS (731.7) and WAIC (719.9) values compared to the interactive model's PSIS (751.6) and WAIC (739.3) values.